Cloud Native Platform Engineering

3 – VPC: Software Defined Datacenters

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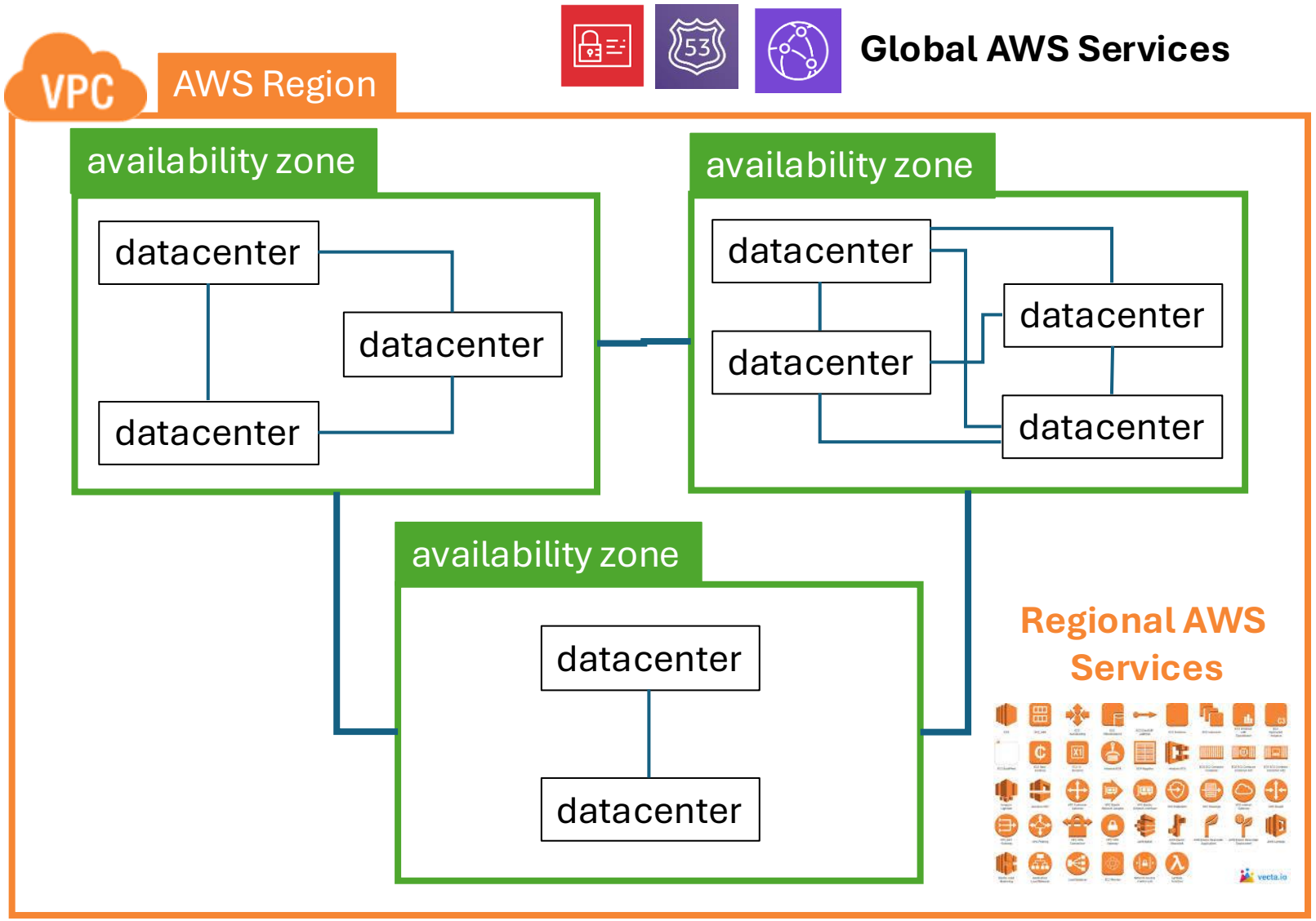
VPCs– Software Datacenter Clusters



- AZs within a region, and datacenters within an AZ are linked together with high-speed, low latency networks
- Most of AWS services are scoped to individual regions, several services are global
- Not every region has every AWS service
- AWS regions provide a clustered platform to enable scalable and resilient application deployment

Properly deploying scalable and resilient applications within a region requires some networking knowledge

IP Addresses

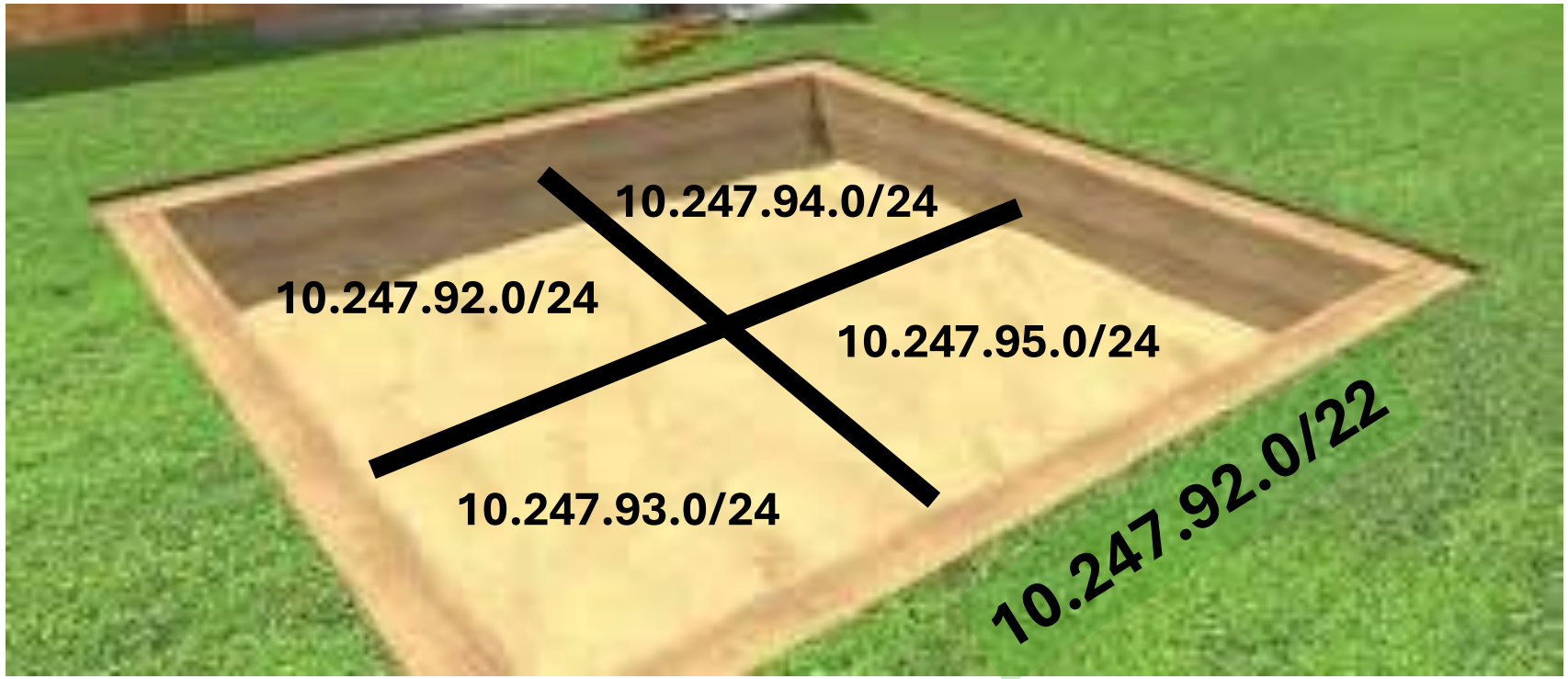
Dotted Decimal	192.168.1.45			
	1st Octet	2nd Octet	3rd Octet	4th Octet
Decimal	192	168	1	45
Binary	11000000	10101000	00000001	00101101
Subnet Mask	11111111	11111111	10000000	00000000
Network Host Split	11000000 10101000 0		0000001 00101101	
Network ID	11000000 10101000 0		0000000 00000000	
Network ID Dotted Decimal	192	168	0	0

Bitwise AND

IP & Mask Notation: 192.168.1.45
255.255.128.0

CIDR Notation: 192.168.1.45/17

Subnetting and “Sandboxes”



00001010	11110111	010111	00	00000000	10.247.92.0/22
00001010	11110111	010111	00	00000000	10.247.92.0/24
00001010	11110111	010111	01	00000000	10.247.93.0/24
00001010	11110111	010111	10	00000000	10.247.94.0/24
00001010	11110111	010111	11	00000000	10.247.95.0/24

Overall network administrator will provide address ranges for a VPC. That is the overall sandbox

You can then use leading bits from the “host” part to further divide up the network

These are called Subnets

Note that AWS reserves 5 IP addresses in every subnet for its own use. Address 0 is the **network identifier**, 1 is the **VPC router**, 2 is the **DNS resolver**, 3 is **reserved for future use**, and the highest address all bits 1 are for the **broadcast address**

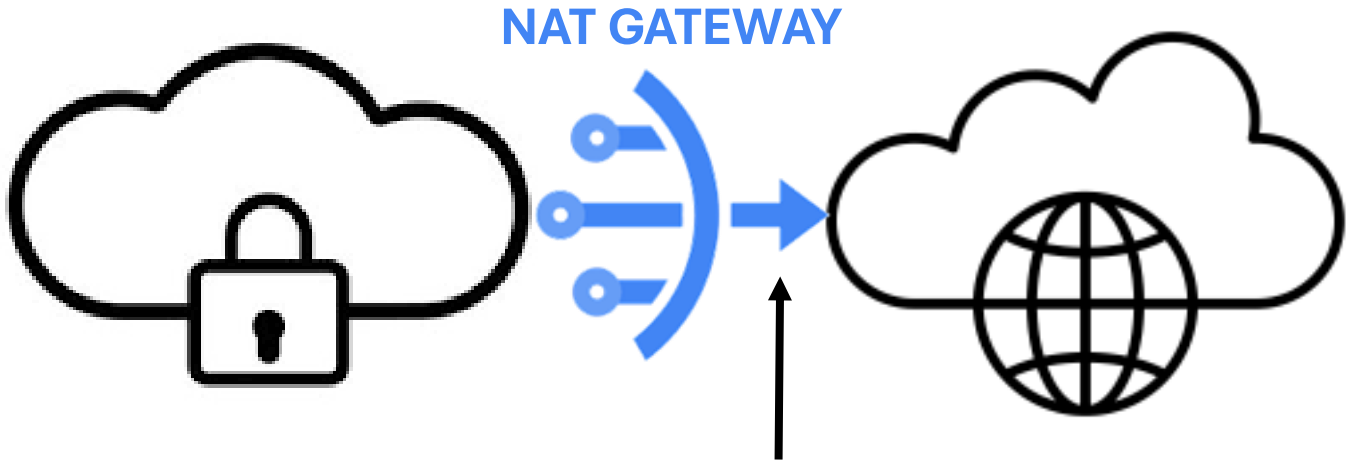
Private and Public Network Addresses

RFC 1918 Defined 3 Private (Non-Routable) address ranges

1. 10.0.0.0/8
2. 172.16.0.0/12
3. 192.168.0.0/16

Any IP addresses in these ranges are considered to be non-routable on the **public internet**. Thus any addresses in these ranges are considered to be on a **private network**

Private IP Addresses in RFC 1918 Ranges. Internet traffic **MUST BE INITIATED** from the private network side



Public IP Addresses NOT in RFC 1918 Ranges, if any traffic with these addresses got to the Internet they would be dropped

Public IP Adress as Proxy for all Private IP traffic ingress and egress to internet

IP Assignment: How Instances Get Their Addresses

DHCP (Dynamic Host Configuration Protocol)

When an EC2 instance launches, AWS acts as a DHCP server — automatically assigning a private IP from the subnet's CIDR range. You can also assign a specific IP at launch. The private IP persists for the lifetime of the instance (even across stop/start).

Private IP Address

- RFC 1918 range (10.x, 172.16.x, 192.168.x)
- Assigned by AWS DHCP at launch
- Stays assigned for instance lifetime
- Survives stop → start
- Never routable on the public internet
- Used for all internal VPC communication

Public IP Address (two types)

Auto-Assigned Public IP

- Free
- Ephemeral — lost on stop
- Changes every restart
- Cannot be predicted
- OK for dev/test

Elastic IP (EIP)

- $0.005/hr if unused
- Persistent — survives stop
- Reassignable to any instance
- Required for stable public endpoint
- Limited: 5 per region by default

⚠ *Stop an instance without an Elastic IP → it gets a new public IP on restart. Scripts or DNS records pointing at the old IP will break.*

Private IPs are stable for the instance lifetime. Public IPs are ephemeral unless you use an Elastic IP — important for any public-facing resource.

Controlling Public (Internet Access)

AWS VPCs are Designed around the Constructs of PUBLIC and PRIVATE Subnets

AWS Internet Gateway

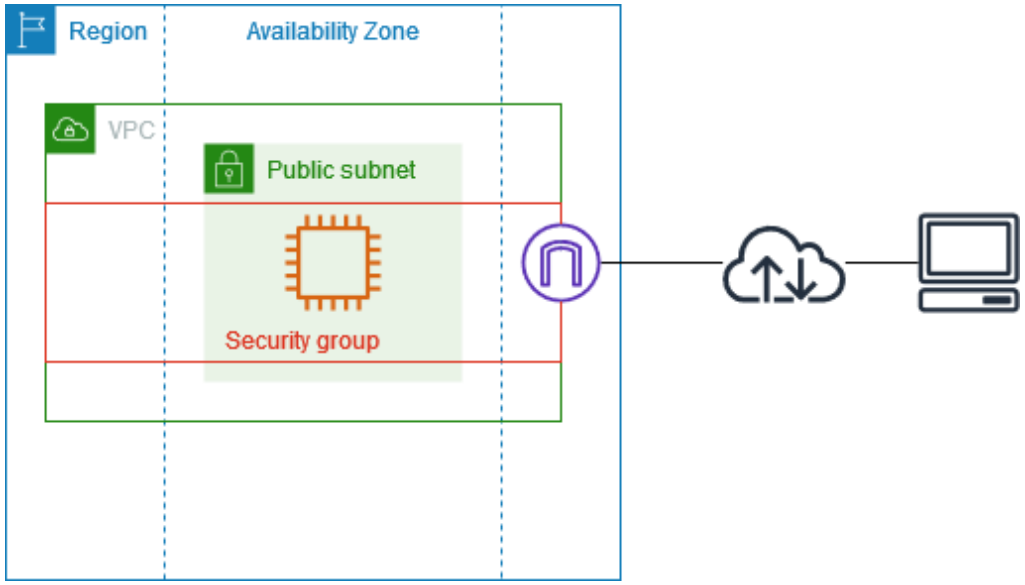
VPCs are Private by default. That means all subnets are private by default - No Internet Access. To gain Internet access an AWS Internet Gateway MUST be attached to the VPC. AWS Internet Gateways have no cost.

AWS NAT Gateway
(Network Address Translation)

Once an Internet Gateway is attached to a VPC, private subnets still have no Internet access. To gain internet access you must deploy add routing to an AWS NAT Gateway service. NAT Gateways have costs about $0.05 per hour (~$1.20/day, ~$36.50/month)

Public and Private Subnets

PUBLIC SUBNET

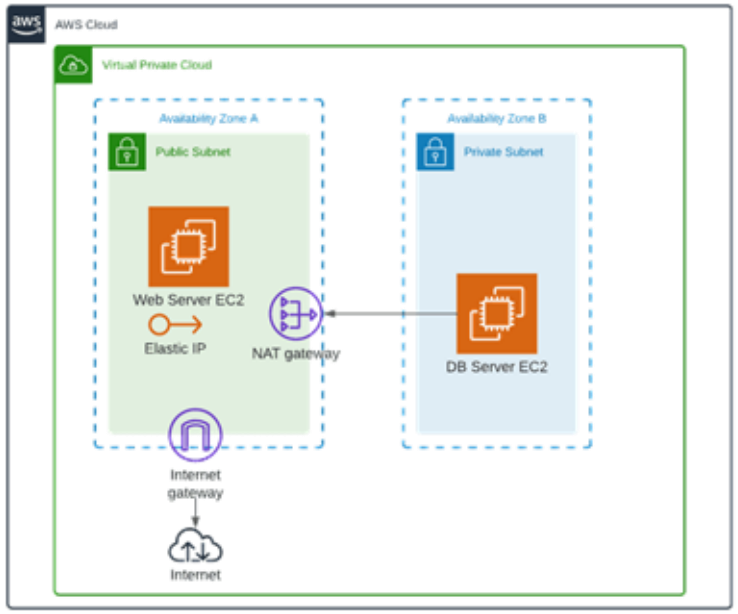


Must have Route Table Entry:

Destination	Target	Status
0.0.0.0/0	ig-xxxxxxx	Active

Internal Resources must have a public IP or an elastic IP assigned

PRIVATE SUBNET



All resources must have private IP addresses (RFC 1918) 10.0.0.0/8, 172.16.0.0/12 192.168.0.0/16

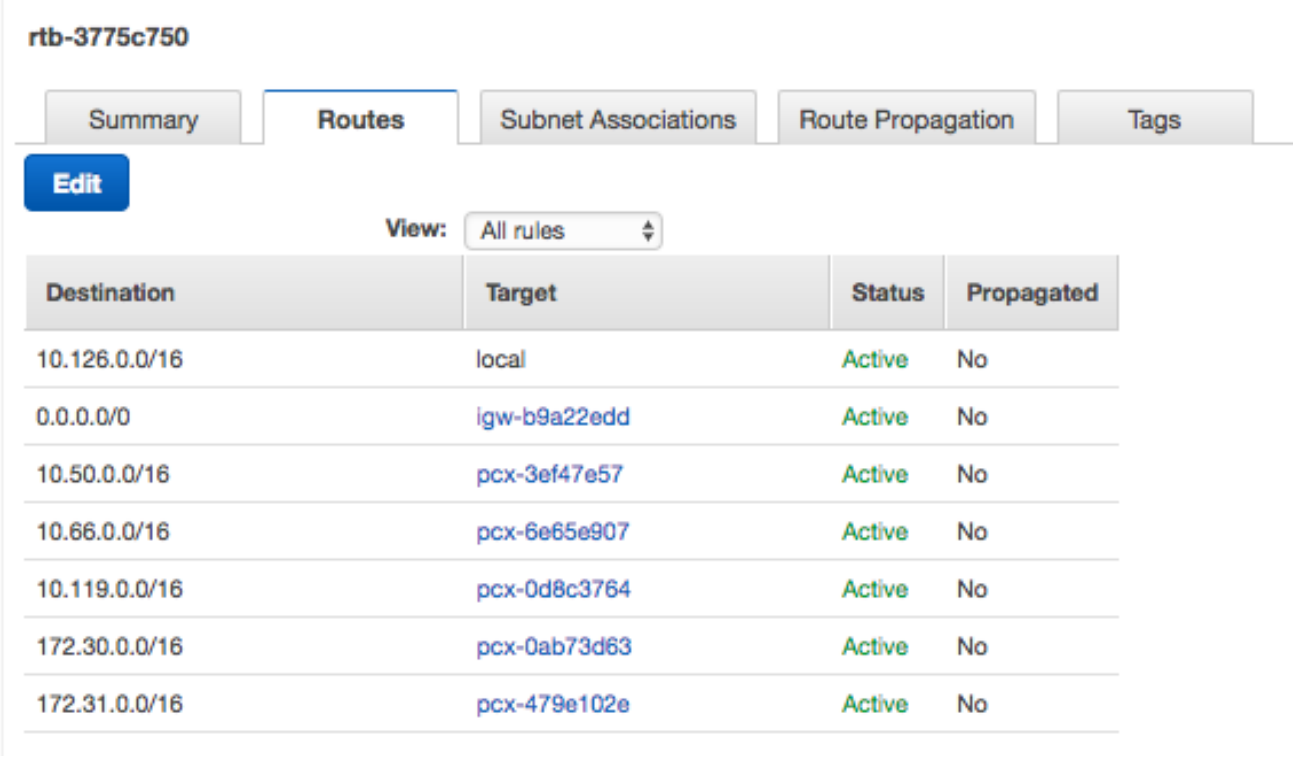
No routes to internet gateway

Internet access only via NAT gateway in public subnet

Destination	Target	Status
0.0.0.0/0	nat-xxxxxxx	Active

Route Tables

"A **route table** is a set of rules used by network devices to determine the best path for directing data packets from their source to a specific destination network."



rtb-3775c750

Summary | Routes | Subnet Associations | Route Propagation | Tags

Edit

View: All rules

Destination	Target	Status	Propagated
10.126.0.0/16	local	Active	No
0.0.0.0/0	igw-b9a22edd	Active	No
10.50.0.0/16	pcx-3ef47e57	Active	No
10.66.0.0/16	pcx-6e65e907	Active	No
10.119.0.0/16	pcx-0d8c3764	Active	No
172.30.0.0/16	pcx-0ab73d63	Active	No
172.31.0.0/16	pcx-479e102e	Active	No

VPCs are configured with a default route table associated with all subnets

Additional Route tables can be created and associated with a VPC

VPC can have MANY route tables

Subnets must be associated with exactly one route table

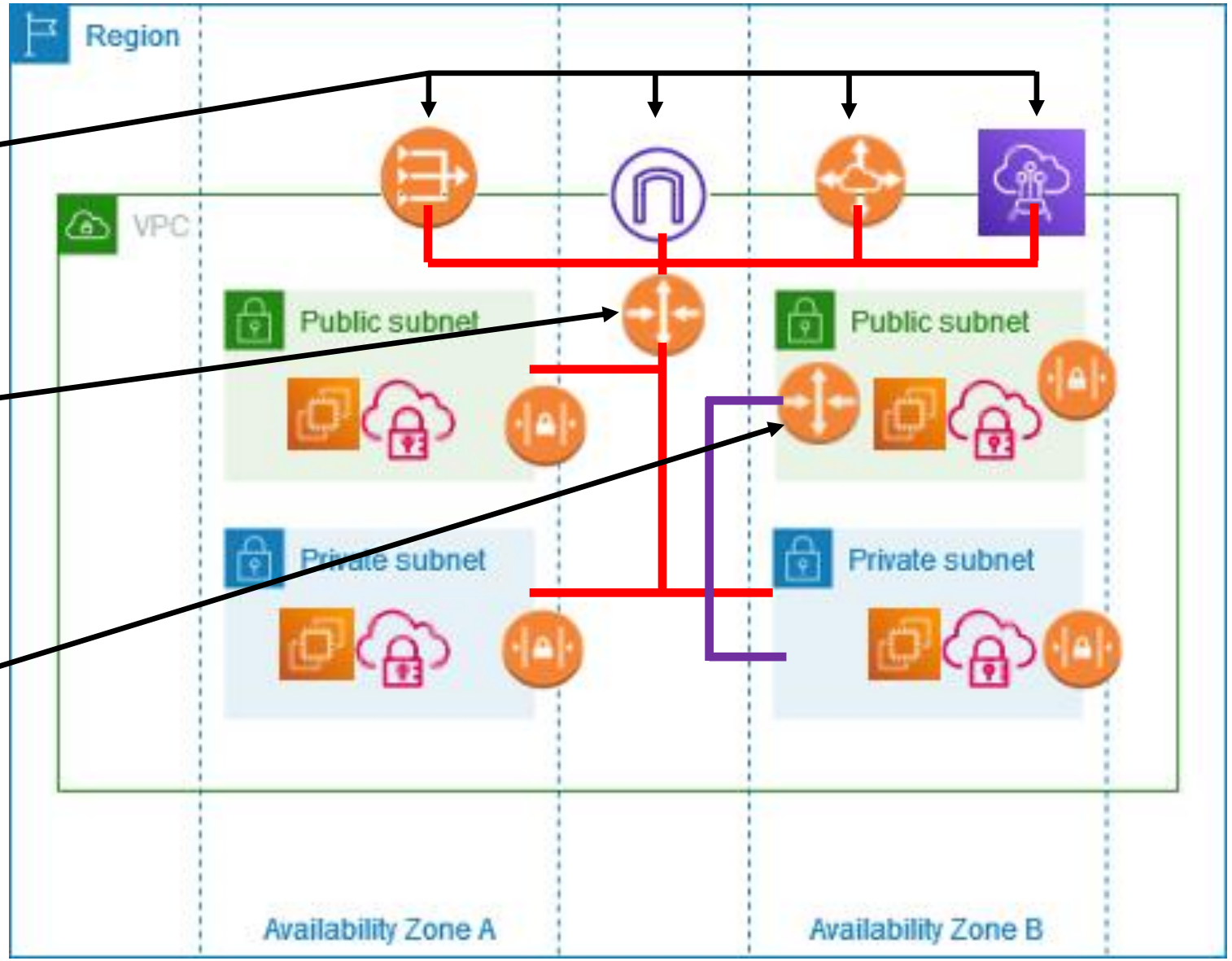
Route targets may be a network CIDR range or be an AWS resource such as an internet gateway

Route Tables

Routing traffic outside the VPC is managed by route tables

All VPCs have a single global route table

Individual subnets may define their own route table to override or add rules found in the global route table



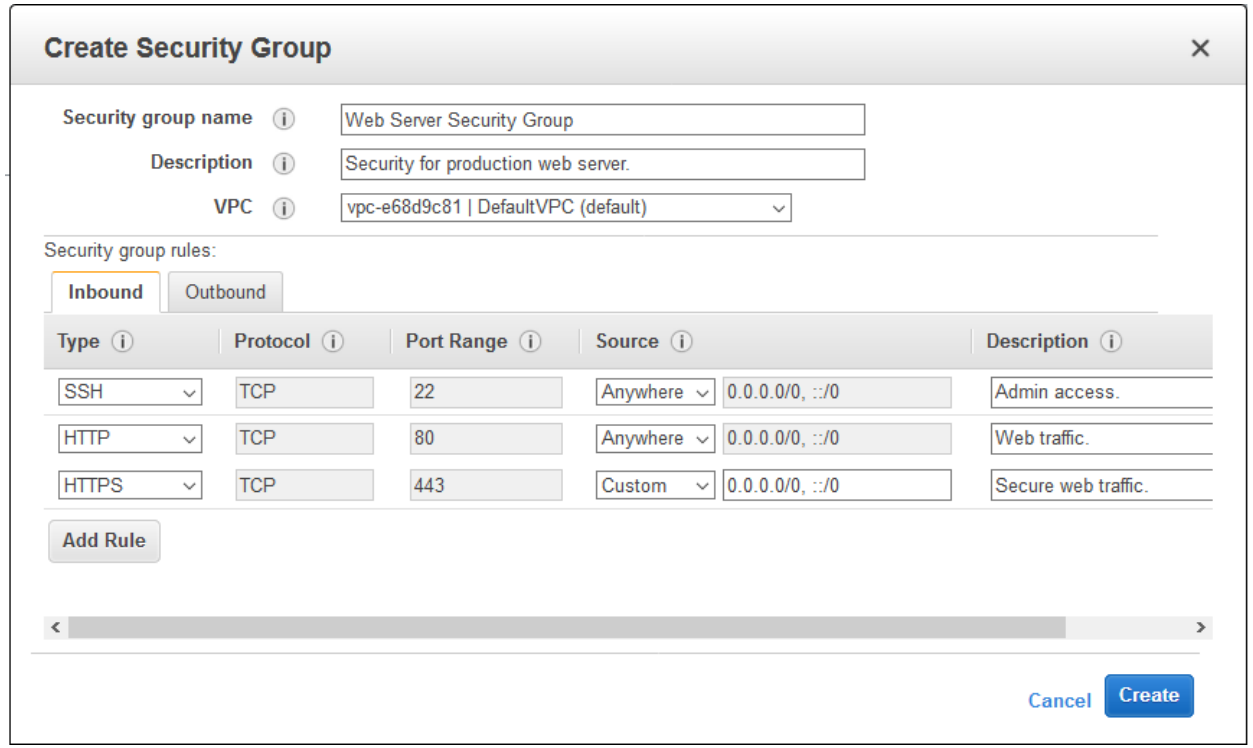
Security Groups

STATEFUL: If traffic is allowed out, return traffic is allowed in and vice versa

ALLOW ONLY: Rules can be allow only, if no rules match, traffic is denied

Rules match based on Protocol, Port, Port Range

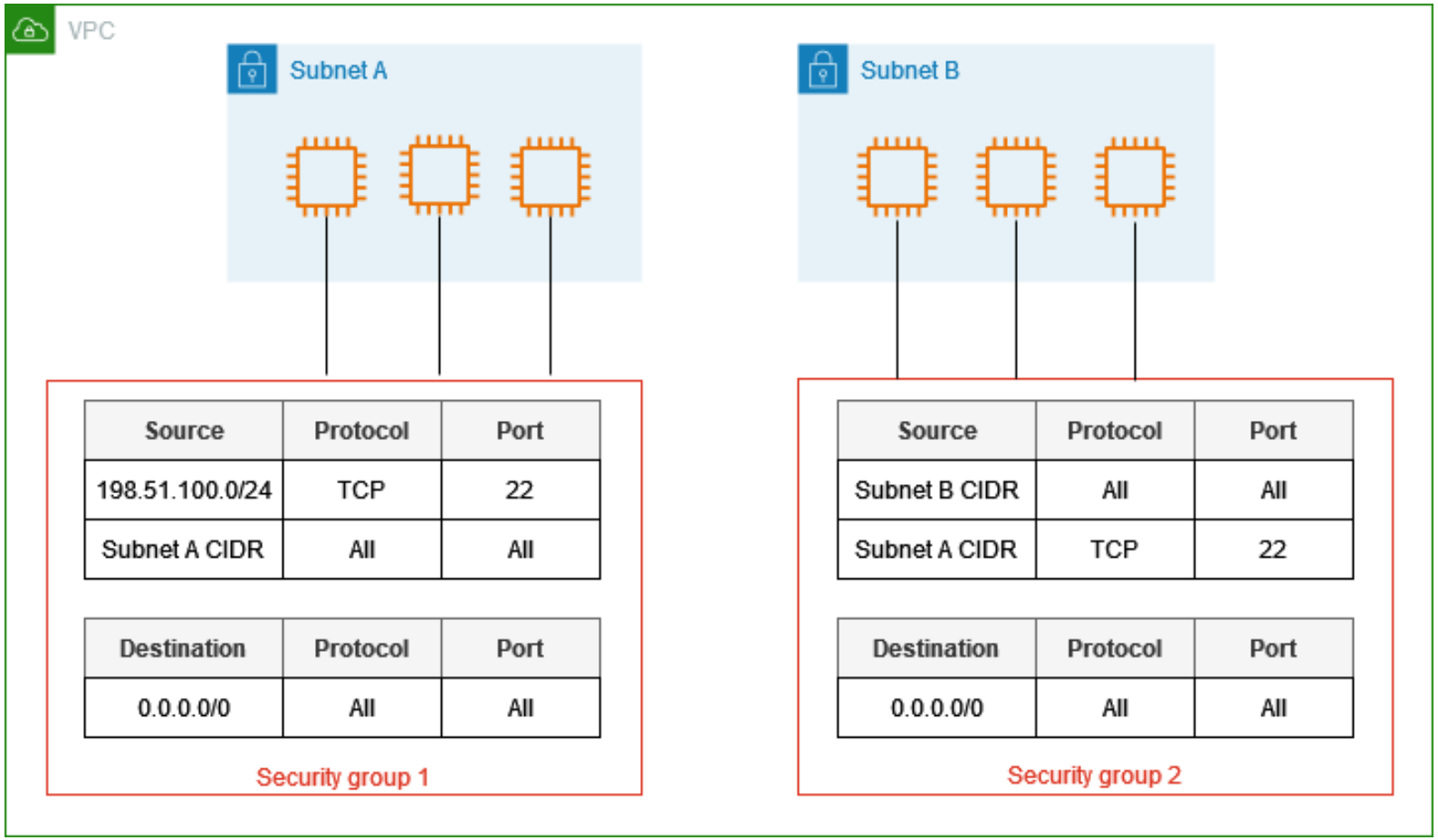
Source can be IP Range (CIDR) or AWS Resource (e.g., Internet Gateway)



Security Groups

Security Groups are the most granular, they are applied to individual resources in a subnet.

Recall NACLs are for entire subnet, and Route Tables are for the entire VPC unless a custom route table is defined for a specific subnet



Source	Protocol	Port
198.51.100.0/24	TCP	22
Subnet A CIDR	All	All

Destination	Protocol	Port
0.0.0.0/0	All	All

Source	Protocol	Port
Subnet B CIDR	All	All
Subnet A CIDR	TCP	22

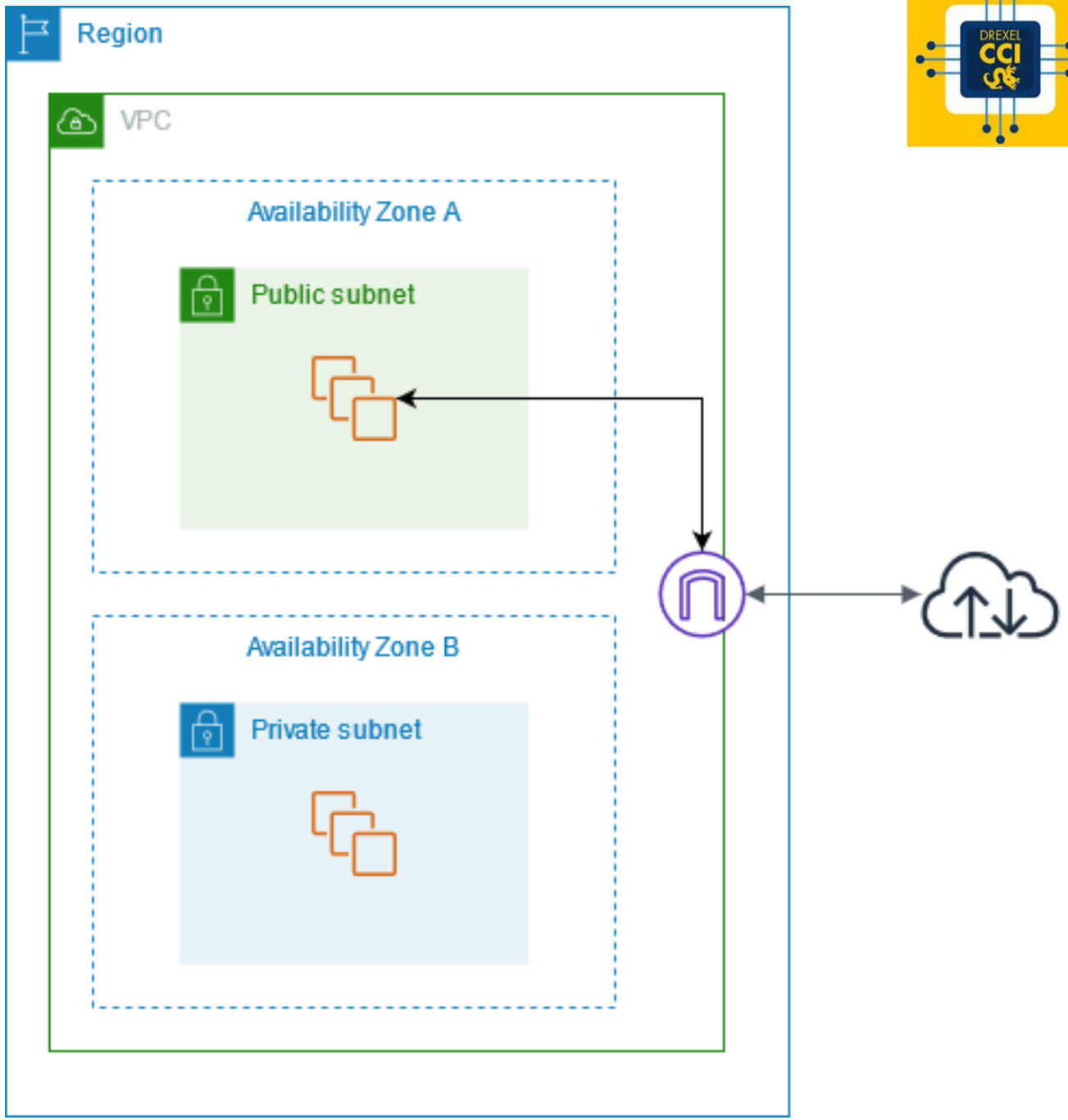
Destination	Protocol	Port
0.0.0.0/0	All	All

Internet Gateway

Connects to edge of VPC that enables internet access

All resources in public subnets with public IPs can send and receive data over internet

Can be restricted with things like security groups

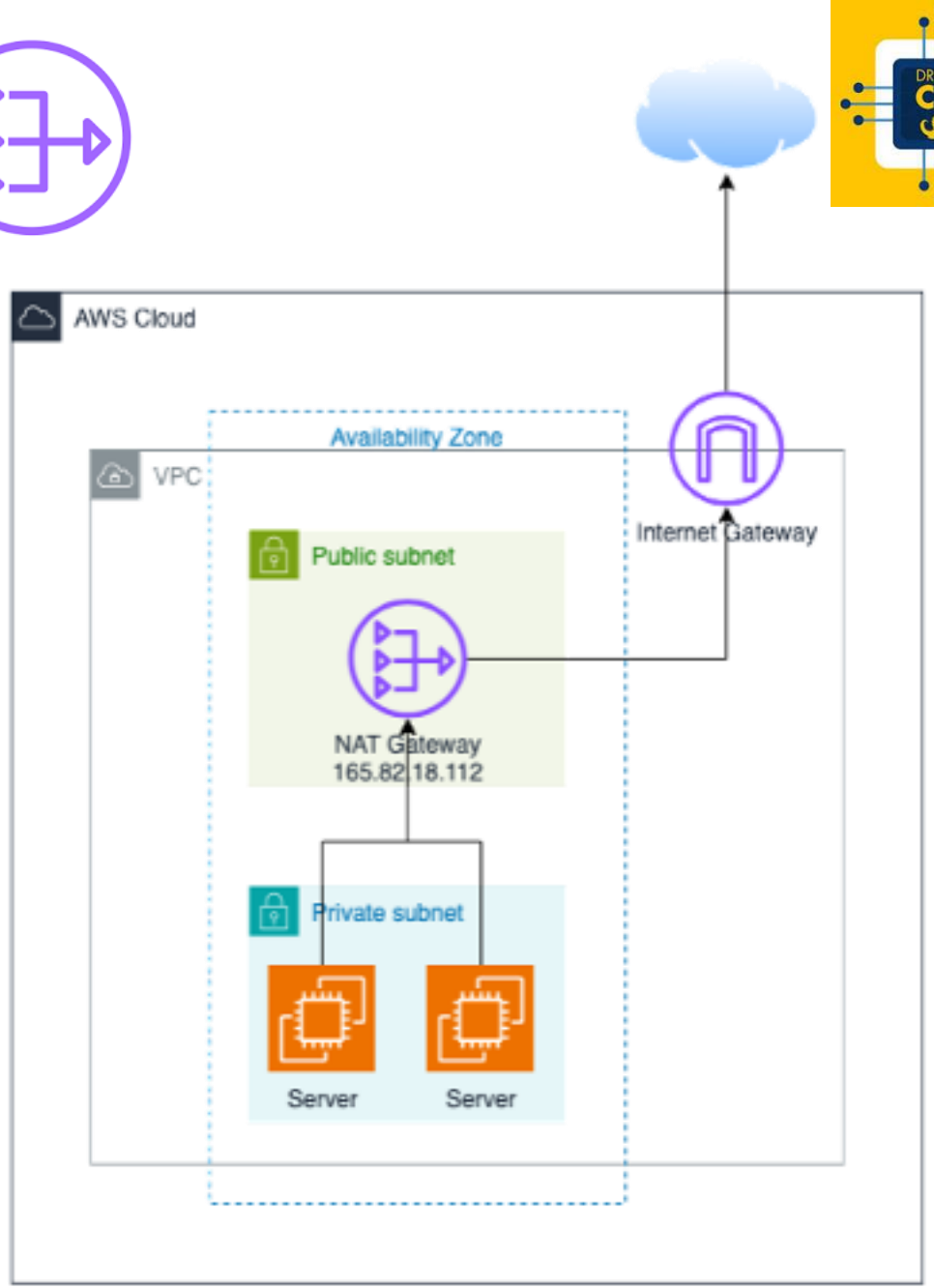


NAT Gateway

RFC 1918 Private addresses are used in a private subnet

Internet traffic targeted at NAT gateway, via NACL

NAT gateway in public subnet

Proxies traffic to and from internet

Internal servers not reachable from Internet